



S.V. National Institute of Technology, Surat

Department of Electronics Engineering

MID Semester Exam, September-2025

B. Tech. (EC) V Semester

Communication Networks (EC321)

Date: 24/09/2025 & Time: 11:15 AM to 12:45 PM

Maximum Marks: 30

Duration: 1:30 Hours

Instructions:

1. [CO] indicates course outcome.
2. All the questions are compulsory.
3. Marks distribution for each question is clearly provided in the 'Marks' column.

Q.1	Write the answers in a few words.	Marks	CO
	a What are the basic components of a data communication system, and what role does each component play?	2	[CO1]
	b Compare the star, bus, and ring network topologies in terms of their advantages and disadvantages in terms of performance parameters.	2	[CO1]
	c Draw the flow chart of ALOHA protocol and explain it.	3	[CO2]
	d How does the CSMA/CD protocol work in Ethernet, and why can't the same collision detection method be used in wireless networks (which use CSMA/CA instead)?	3	[CO4]
	e Explain Go-back-N protocol with proper timing diagrams and sliding window diagrams.	3	[CO2]
	f Explain Packet delivery ratio, Throughput and Latency and its components.	3	[CO1]
	g Differentiate between a physical address, an IP address, and a port address in the context of networking. What is each one used for?	4	[CO1]
Q.2	Solve the following Questions.		
	a Explain the concept of layering in data communication networks and describe the functions of each layer of the OSI model.	5	[CO2]
	b In case of CDMA, design a Walsh table for a network having 6 Nodes (Let the names of the nodes be A, B, C, D, E and F) (2 Marks). Assign one code to each node. Node A is transmitting bit '1', B is transmitting bit '0' and C is transmitting bit '1' and Node D is receiving. Show how D can detect the bits transmitted by A, B and C. (1+1+1 Marks) (Only theoretical explanation is not sufficient. Draw the appropriate codes/data/matrix/graph.)	5	[CO1]
